



SAS[®] Viya[®] Platform Operations: Logging

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Overview

Monitoring log messages is an important responsibility for a Kubernetes administrator. By monitoring log messages, administrators can understand the health and performance of their cluster and the applications running on it.

There are different ways of monitoring log messages. You might use established tools and processes for log monitoring. If so, SAS Viya platform components follow cloud-standard patterns when producing log messages, which should make it easier to use your existing log monitoring tools to monitor SAS Viya platform. However, if you do not have an established preferred log-monitoring tool, here are additional options for viewing log messages:

- Use the SAS Viya Monitoring for Kubernetes solution. You can deploy this solution, without charge, from the [SAS Viya Monitoring for Kubernetes](#) GitHub site.

For more information, see the [SAS Viya Monitoring for Kubernetes](#) documentation.

- Use the `kubectl` command to review log messages that are generated by the applications and pods in their clusters.

For more information, see [“Use kubectl Commands to View Log Messages” on page 2](#).

How To

Use kubectl Commands to View Log Messages

List All Pods

To display log messages, you must specify the pod. Use the following command to obtain the pod names in your environment.

```
kubectl -n name_of_namespace get pods
```

Where *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.

The following output is an example of the information returned:

NAME	READY	STATUS	RESTARTS	AGE
annotations-6cd7f58866-55wp4	1/1	Running	2	31h
appregistry-58f8f7fc8d-4zqts	1/1	Running	0	31h
arke-58d548b47d-p6zm4	1/1	Running	7	31h
audit-cc5f46f9d-pkrj7	1/1	Running	12	31h
authorization-8db874d94-l2rbc	1/1	Running	0	31h
cachelocator-664cd5777d-57mcb	1/1	Running	0	31h
cacheserver-85b658c5bf-nqts4	1/1	Running	0	31h
cas-control-56978d9c79-8z7c9	1/1	Running	8	31h
casadministration-5d69cd6f9d-6mj5d	1/1	Running	0	31h
casoperator-8d6fdd496-d9hcs	1/1	Running	0	31h

Display the Log Messages for a Specific Pod

```
kubectl -n name_of_namespace logs pod_name
```

Where:

- *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.
- *pod_name* is the pod for which you want log messages returned.

Display the Log Messages for a Specific Container within a Pod

```
kubectl -n name_of_namespace logs pod_name -c container_name
```

Where:

- *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.
- *pod_name* is the pod for which you want log messages returned.
- *container_name* is the container for which you want log messages returned. The *container_name* is the *pod_name* without the trailing numbers.

Note: Use the `--previous` option on the above command to display the previous container's log. This is useful if the container is in the process of deploying and you do not want to wait for it to fail (again).

View Log Messages for a Specific Amount of Time

```
kubectl -n name_of_namespace logs --since=time pod_name
```

Where:

- *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.
- *time* is the time at which to start the return of the log messages.

The variable can be expressed in the following ways:

- ☐ *secs*
- ☐ *minm*
- ☐ *hourh*

For example, `--since=3h` specifies to return log messages for only the last three hours.

If you use the `--since` option, you cannot also use the `--since-time` option.

- *pod_name* is the pod for which you want log messages returned.

View a Specified Number of the Most Recent Log Messages

```
kubectl -n name_of_namespace logs pod_name --tail=number_of_messages
```

Where:

- *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.
- *pod_name* is the pod for which you want log messages returned.
- *number_of_messages* is the number of log messages that you want returned.

View Log Messages since a Specified Date

```
kubectl -n name_of_namespace logs --since-time=date pod_name
```

Where:

- *name_of_namespace* is the namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.
- *date* specifies the date and time (in RFC3339 format) after which logs are returned. See [Date and Time on the Internet: Timestamps](#) for information about the RFC3339 format.

If you use the `--since-time` option, you cannot also use the `--since` option.

- *pod_name* is the pod for which you want log messages returned.

Use kubectl Commands to View Kubernetes CronJobs and Jobs

Note: To perform this task, you must have elevated Kubernetes permissions.

Overview

Many components of the SAS Viya platform use Kubernetes CronJobs. Here are three examples:

- the SAS Viya backup and restore utility. For information, see [SAS Viya Platform: Backup and Restore](#).
- CAS table state management, which enables you to manage the import, load, and unload of source files in CAS. For information, see [SAS Viya Platform: Data](#).
- Starting at 2025.02, SAS Risk Cirrus solutions deploy as CronJobs, which avoids having a solution pod that runs continuously. For information, see your risk solution's Administrator's Guide (access key required).

A CronJob creates Jobs, which represent one-off tasks that run to completion and then stop. For the list of CronJobs in your SAS Viya deployment, run the following `kubectl` command:

```
export viya=name_of_namespace
kubectl -n $viya get cronjobs
```

For more information, see Kubernetes documentation on [CronJobs](#) and [Jobs](#) resources.

For information about the cron schedule expressions, see [crontab guru](#).

View Job Completion Status

Use the following command to show job status for SAS Viya pods. The output is sorted by the creation time stamp with this option: `--sort-by=.metadata.creationTimestamp`

```
export viya=name_of_namespace
kubectl -n $viya get jobs --sort-by=.metadata.creationTimestamp
```

The completions column shows the status of the job:

- 0/1 = in-progress or failed
- 1/1 = completed

Use the following version of the command to display the `STATUS` column:

```
kubectl -n $viya get jobs --sort-by=.metadata.creationTimestamp -o custom-
columns='NAME:.metadata.name,CREATED:.metadata.creationTimestamp,
STATUS:.status.conditions[0].type'
```

View Pod Logs from a Job

Find the pod logs under the job that you want to inspect.

- 1 Find the pods that are associated with a job:

```
export viya=name_of_namespace
kubectl -n $viya get pods -l job-name=name_of_job
```

- 2 Choose a pod name from the list of pods that were returned and run the following command:

```
kubectl -n $viya logs name_of_pod
```

Access and View Audit-Service Loggers

TIP These loggers display the audit-service log messages in the Kubernetes pod logs (Stdout).

The `SAS_LOG_LEVEL_SERVICE_AUDIT` logger is used by SAS Viya platform audit service. This logger displays logs that are related to all of the log messages in the audit service.

- To ensure that the `SAS_LOG_LEVEL_SERVICE_AUDIT` logger is set to `INFO`. Enter the following command:

```
kubectl -n namespace set env deployment sas-audit SAS_LOG_LEVEL_SERVICE_AUDIT=INFO
```

Where *namespace* is the SAS Viya platform namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.

- To display all audit-service transactions, enter the following command:

```
kubectl -n namespace set env deployment sas-audit SAS_LOG_LEVEL_SERVICE_AUDIT=DEBUG
```

Where *namespace* is the SAS Viya platform namespace for which you want the log messages returned. If you do not specify this option, the current namespace is used.

When this command is issued, a rolling update is applied to the audit service and a new pod is started. The pod generates debug-level log messages. These messages can report the following information:

- ☐ when audit records are uploaded to the database
- ☐ when audit records are fetched or read from the database
- ☐ when operations are performed with CAS or PostgreSQL

See Also

- [SAS Viya: Auditing](#)
- [“Threshold-Level Definitions” in SAS Viya Platform: Log Monitoring](#)